Real-Time Video Inference on 8× NVIDIA RTX PRO 5000

MiniMax H3 with vLLM-Omni

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vLLM-Omni Extends vLLM Beyond Autoregression

One serving layer spans omni-modality inputs, multiple execution loops, and heterogeneous outputs

WHAT vLLM-OMNI ADDS

01 Omni-modality

Text, image, audio, and video enter one serving pipeline

02 Multiple execution loops

AR decoding and DiT denoising follow different schedules

03 Heterogeneous outputs

One request may emit text, image, audio, video, or actions

ONE ARCHITECTURE



STAGE GRAPH placement · batching · transport · lifecycle

Each stage owns its execution policy

MINIMAX H3 Text Encoder → DiT → VAE Decoder

Logical components stay co-located inside one diffusion stage

Source: vLLM-Omni v0.26 public deck, slides 4 and 22–24 · architecture overview · arXiv:2602.02204

SM120: Compute and Communication Regimes

Baseline: 8× NVIDIA RTX PRO 5000 72GB · compute capability 12.0 · 110 SM per GPU

sm_120

CUDA compute capability 12.0

110
SM / GPU*

72 GB
GDDR7 ECC / GPU

1,344 GB/s
local memory BW / GPU

PCIe 5.0 ×16
system interface

8 GPUs **880 SM · 576 GB**
aggregate sums · memory is not pooled

*14,080 CUDA cores ÷ 128 FP32 cores per CC 12.0 SM

CLASSIFY BY TENSOR OWNERSHIP

COMMUNICATION-MODEST

PLACEMENT tensors remain local

SYNC few readiness fences

EXAMPLES projection · MLP · local VSA tiles

COMMUNICATION-INTENSIVE

PLACEMENT Q / K / V / Gate / O change rank

SYNC barriers sit on the dependency path

EXAMPLE Ulysses SP8 attention



increasing remote bytes + synchronization frequency

1,344 GB/s
local GDDR7

vs **50 GB/s**
nominal 400 Gb/s port

26.9× raw-rate ratio*

8 active 400 Gb/s ports · *local memory bandwidth ÷ nominal NIC line rate; not end-to-end payload throughput

Sources: NVIDIA RTX PRO 5000 Blackwell datasheet · CUDA GPU Compute Capability · CUDA C Programming Guide · measured system inventory